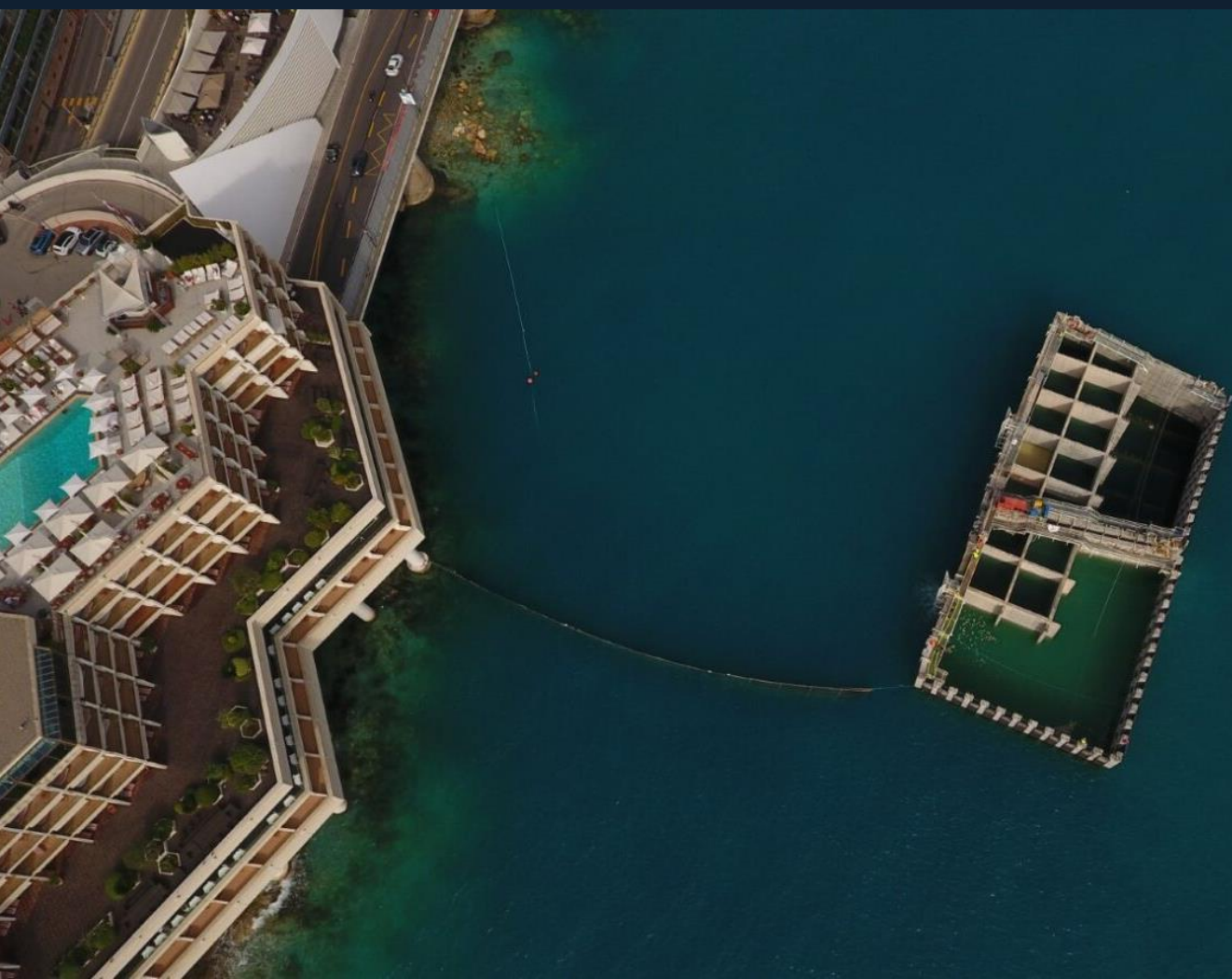


Health & Safety

Standard River and maritime works



Reference

Company	Document Type	Department	N°	Version	Langue
BYTP	REF	H&S	2229	A	EN

Version	Date	Written by	Checked by	Approved	Reason for modification
A	01-11-2023	M.RABAUD	I.LELIEVRE	L.KNOLL	First issue

The standards presented in this document aim to provide a practical guide to define the means to be deployed for the river and maritime activities.

Table of contents

- 1. Work surfaces
- 2. Life jackets
- 3. Boarding stations
- 4. Access to workstations
- 5. Boats
- 6. Floating devices
- 7. Retaining walls methods (hammering, vibrator, jacking)
- 8. Emergency organisation
- 9. Underwater work



1. Work surfaces

- Stable surface (boom, decking, piling guide, etc.) = **rigid guardrails on the periphery**
- Lifebuoy with 30 m rope and placed at a maximum distance of 50 m from each other on all stations
- Use of life jacket based on site conditions and specific risk analysis
- Floating equipment (boats, pontoons, barges, etc.) = **wearing a life jacket compulsory**
- Rigid guardrails to be adapted to the environment and the risk analysis of the project
- Equip the floating device following the instructions on the registration certificate
- Personnel working on floating equipment must know how to swim a short distance (ideally with a swimming certificate)



2. Life jackets

- Define the instructions and carry out a practical demonstration at the safety induction
- Enure personnel awareness of the visual inspection of the life jacket before its use
- Ensure an yearly inspection by a certified organization
- Adapt the life jacket to to the risks identified (work in hot spots = fireproof vest)



Hydrostatic vest with automatic release



Manual release vest



Foam vest (non-production staff)



Fireproof life vest



Life jacket/harness for working at height

- Prioritise the wearing of automatically released vests starting from 150 N (newtons)
- Drivers of machines working on floating pontoons (excavator, crane, drill, etc.) must be equipped with manual release vest
- Perform a visual check before each use (cartridge, sewing, damage, etc.)
- For nightshift, equip the vest with a flashlight

VIGILANT POINT

- The use of a life jacket is mandatory during the assembly and dismantling of temporary structures above water
- The subcutal strap must be tight to avoid any snagging hazard
- Any triggered vest is sent for repair by a specialized company (CO2 cartridge change, vest checks, etc.)

3. Boarding stations

The boarding stations are equipped with:

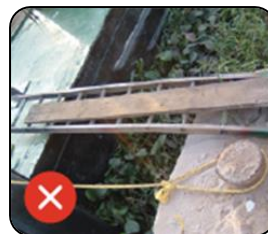
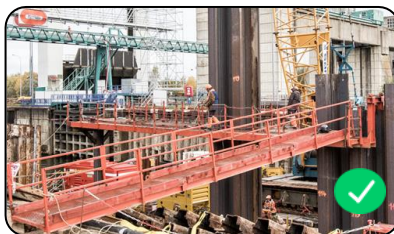
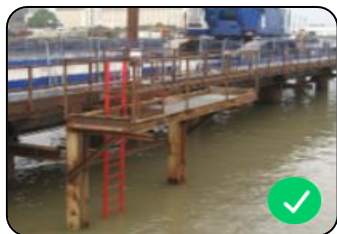
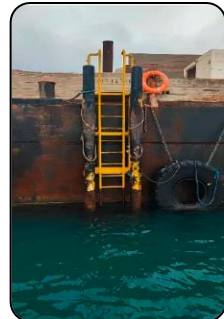
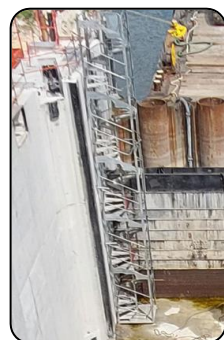
- Support points at hand height
- Non-slip floor
- Access ladder
- Lighting means
- Mooring bollards
- Lifebuoy(s)
- Maritime beacons or lights to guide boats in bad weather (reduced visibility)



4. Access to the workstation

Nautical and/or land access to workstations

- Collect information regarding drafts
- Non-slip step surfaces
- Stairs with more than three steps are provided with handrails
- Metal walkway (fixed or hinged) on the floating craft are equipped with handrails on both sides
- Access to the bottom of the cofferdam to be favored by access tower
- Ladder (sealed and bolted) on pontoon with draft ≥ 1 m
- Access to the hammering guide:
 - Crinoline ladder only by nautical route
 - Access walkway blocked at the feet and fixed at the head by land
- Access at sea: Favor access via boat landing or quay when possible
- Depending on the work environment, provide a remote base on a pontoon/barge



VIGILANT POINT

- The use of a dock ladder must be done with full use of all four limbs = do not carry any tools in the hands and maintain 3 points of contact
- Take into account the impact of tides which can reduce the draft
- Prohibition on jumping ashore when the boat is moving

5. Boats

The boats

- Identified by a readable name (identifier)
- Conducted by designated and authorized personnel
- Boat license required for motorization greater than 4.5kW (6 hp)
- Choose the engine suited to the most critical navigation conditions expected on site (wind, swell, current, etc.)
- Maximum load (kg) and maximum crew (pilot included) indicated on the boat
- The waterline (demarcation between submerged and emerged parts of a ship's hull) is clearly visible and marked with paint
- Maneuver: avoid passing over cables and moorings



Boat equipment

- 1 Pike pole
- 1 lifebuoy with an at least 30m throw line
- 2 oars
- Motor(s) adapted to the current
- Mooring line
- One or more adjustable spotlights in case of intervention at night or in case of fog
- A powder fire extinguisher (in case of an oil fire on water)
- portable VHF

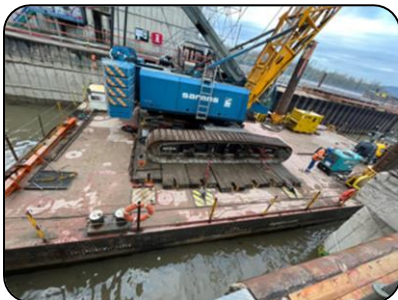


VIGILANT POINT

- For the navigation of boats, barges and pontoon-mounted vehicles, obtain the navigation notice from the harbor master's office, including for signal lights
- Take into account the impact of tides which can reduce the draft at the quay
- Sea/river limit: Consult local regulations to find out the limits of maritime zones and inland water zones (impact on the choice of boat license)
- Keep walkways free of obstacle on board
- Move without rushing, especially when boarding/disembarking
- Height difference between boat and floating device = use of a pilot ladder

6. Floating devices

- Freeboard height ≥ 1 m (distance between the water level and the upper part of the bridge) = Installation of peripheral guardrails
- Freeboard height < 1 m = collective protection according to risk analysis
- Presence of a crane or any other machine on pontoon = stability study
- Armament plan for floating devices:
 - Method statement
 - Limitation of movement for machines
 - Lashing of equipment when required (excavator, lattice crane, etc.)
 - Definition of storage areas and personnel movement areas
- Winches must be equipped with disengagement devices and ensure proper qualification training for its use
- Signalisation of hazardous areas with paint (passage of towing/hauling cables, counterweights motion area, mooring weights, bollards, etc.)
- Intervention in boat holds is subject to a risk analysis specific to confined spaces (Confined Spaces training, permit to work, ventilation or forced ventilation, gas detectors with alarm, pre-entry atmospheric test, PPE, instructions in the event of emergency)
- Mooring stations must be defined in a mooring plan and a specific risk analysis.
- Periodic check of the hawsers and fairleads' conditions.



VIGILANT POINT:

- The angle of heel (tilt) of the boat must not exceed 3° , beyond that, refer to the stability note
- Take into consideration the overturning moment of the pontoon which can be more restrictive than the lifting capacity of the machine
- Adapt the moorings according to the size of the pontoon and the conditions of use (mooring break = pontoon drift and whip effect)
- Prioritise the use of portable power tools
- Anti-pollution kits with floating tubes available on board
- Provide fire extinguishers with anti-rust/anti-corrosion treatment (MED Certificate or equivalent)

7. Retaining walls methods (hammering, vibrator, jacking)

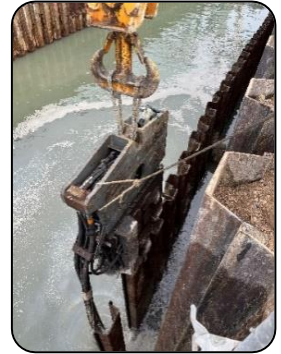
Hammering sheet piles/pipes

- Operators must be trained and authorized to use the equipment
- Wear ear-muffs and earplugs (preferably molded earplugs)
- Equip the pile guide frame with closing hatches and guardrails
- Crimping guardrail devices
- Marking of bolts on hammer and vibratory hammer to monitor potential loosening
- Visual tightening check before each operation
- Keep a minimum distance of 3 m from the vibratory hammer / hammer (do not stay under it)
- Carry out daily checks of the equipment (crack detection, particularly during winter)
- Provide permanent slings for hydraulic power stations



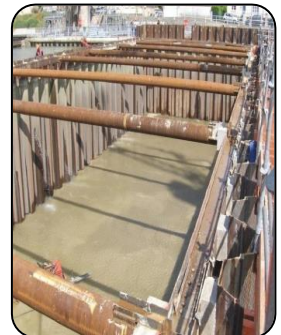
Tearing off tubes/sheet piles

- Check the compatibility of the material with the required traction force
- Visual check of the full sheet pile / tube during the tearing off
- Before any operation to release the vibratory driver, the operator must have received the validation from the crane operator
- Secure the deposit perimeter and provide a storage device



Cofferdams

- Sizing of the Liernes (Vault)/buttons carried out by the technical department
- Ensure proper lifting points on the liernes and buttons
- Reception of cofferdams by the Temporary Works Engineer
- Removal must be part of the installation procedure



VIGILANT POINT

- Check the welding points of the hatches/guardrails/access (risk of damage due to strong vibrations)
- Storage of sheet piles / tubes in a barge or on the ground = guarantee stability at all times, free and clear access around, as well as easy access to hook the lifting device
- Drain pumps must be positioned to be accessible at all time and to not obstruct the circulation

8. Emergency organisation

In case of fall into water

- A notice board on site, in a prominent location, must indicate the rescue instructions
- Personnel must be train for aquatic risks (drowning, hydrocution and hypothermia, buoy throwing, man in water, etc.)
- The length of the lifebuoy rope must be adapted to variations in tides and height of structures (minimum 30m)
- Rescue boat = License + authorization by project management + safety equipment + VHF radio
- Always at least 2 people on the rescue boat to carry out the rescue
- Mandatory safety equipment on board a lifeboat (at a minimum):



Safety jacket



Life buoy + throw line



Pike pole



Rescue and emergency knife



Towing and mooring device

- Additional rescue devices: fire extinguisher, first aid kit, lighting device for research, flashlight for emergency personnel
- Alarm device on the boat or fog horn



Use of an emergency basket for rescues:

- Formalize a specific procedure and external emergency services
- Ensure it is knoww by the rescuers, the crane operator and the project management
- This pod is exclusively dedicated to the evacuation of an injured person

VIGILANT POINT

- Check that the rope is securely attached to the buoy and is not tangled
- Be careful of rope knots and that the rope is properly attached to the buoy
- Carry out regular buoy throwing training
- Train relevant personnel to escape from a cabin in the event of a fall into water
- Turn off/neutral the engine when approaching the victim, risk of being struck by the propellers (cage around the propeller)
- Present yourself in a position adapted to the current and wind
- Once recover onboard and maintain the victim in a horizontal position rather than vertical position
- Wrap the victim with dry clothing and survival blanket



9. Underwater work

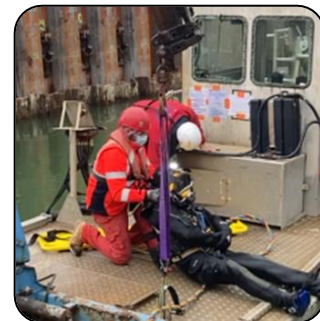
Underwater work intrinsically induces a risk linked to human exposure and its technical requirements. These operations should out at least be carried according to the following requirements:

- Divers must comply with applicable local and international regulations
- Operations must be subcontracted only to accredited companies guaranteeing qualified/ competent personnel for the diving and the operations to be carried out.
- All diving operations are subject to a specific risk analysis validated by the Diving Supervisor
- Specific procedures are prepared based on these risk analyses
- A specific emergency plan for these operations is drawn up
- A diving team is at least composed of:
 - 1 Diving Supervisor
 - 1 Diver
 - 1 hyperbaric operator (rescue, operator assistance) whose role is defined according to the complexity of the operations to be carried out, the emergency means available on site and the risk analysis which can define additional human resources
- Permanent radio communication is maintained between the diver(s) and the Diving Supervisor
- The equipment must be confirmed in good condition and compliant with the operations to be carried out considering the environment (sewage treatment plant, confined environment, ionizing environment, etc.)
- Diving work is suspended as soon as the conditions are likely to endanger the operator (climate change, unforeseen coactivity particularly in superposition, etc.)



Co-activity management

- Visual materialization of the diver's area (at water level for watercraft and at height if required for local cranes)
- Any lifting maneuvering or movement in the diving area must be validated by the Diving Supervisor who will remove his team from the water if necessary
- The guiding of parts in water is carried out by the Diving Supervisor
- The diver must never be located between a moving element and a fixed surface (load to be lifted, river craft in its work area, etc.)
- The overlapping of activities in water / out of water is prohibited
- The hyperbaric team must be positioned as close as possible to the diver



The difference in pressure between the sides of a structure/equipment creates a suction phenomenon which can cause the diver or their equipment to become trapped (major bodily danger)

Water movements near the diver must be validated by the Diving Supervisor